

Permutations and Combinations

COMP2711 Discrete Mathematical Tools

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SECTION 1

Counting models

Motivating Example: Selecting three people to compete in ICPC

Choose three students from a group of six to form an ICPC team.



How many different teams can we form?

Team Red Bird Flies First at ICPC Asia East
Continent Final in Hangzhou (Junhe ZHENG, Kaibin
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Order is ignored: **a combination**.

ICPC team photograph

Choose three of the same six students, then arrange them for a photograph.



How many different photograph arrangements are possible?

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Swapping two people changes their left-to-right order:

$$(A, B, C) \neq (B, A, C).$$

Order matters: **a permutation.**

Choose the model

MUST UNDERSTAND Ask whether swapping two selected objects changes the outcome.

Permutation	Combination

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example	arranging photographs	selecting ICPC teams

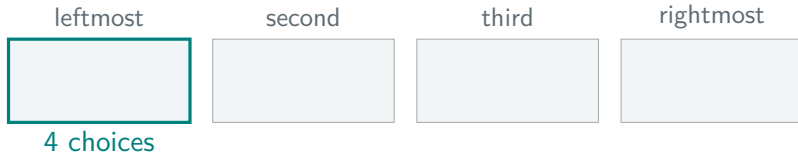
SECTION 2

Permutations

Four people in a photograph

Arrange A, B, C, and D from left to right.

How many different photographs are possible?



Available: **A, B, C, D.**

For example, choose B for the leftmost position.

Four people in a photograph

Arrange A, B, C, and D from left to right.

How many different photographs are possible?



Remaining: **A, C, D**.

Choose C for the second position. B cannot be chosen again.

Four people in a photograph

Arrange A, B, C, and D from left to right.

How many different photographs are possible?



Remaining: **A, D**.

Choose A for the third position.

Four people in a photograph

Arrange A, B, C, and D from left to right.

How many different photographs are possible?



Remaining: **D**.

D takes the last position.

Four people in a photograph

Arrange A, B, C, and D from left to right.

How many different photographs are possible?



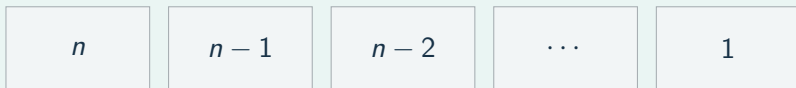
By the product rule,

$$4 \cdot 3 \cdot 2 \cdot 1 = \boxed{24 \text{ photographs}}.$$

Factorials

For four people, we counted $4 \cdot 3 \cdot 2 \cdot 1 = 24$ arrangements.

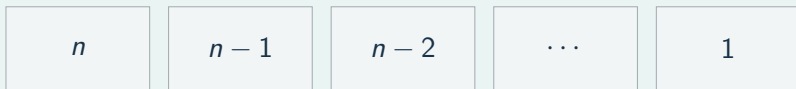
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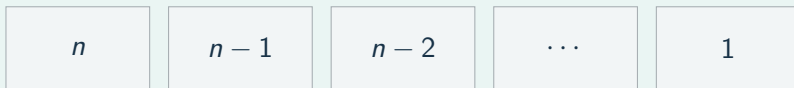


$$n! = n(n - 1)(n - 2) \cdots 2 \cdot 1$$

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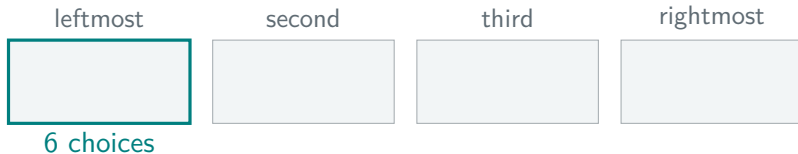


$$n! = n(n - 1)(n - 2) \cdots 2 \cdot 1$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24 \quad \text{and by definition} \quad \mathbf{0! = 1.}$$

Four positions, six people

Choose four of A, B, C, D, E, and F for a photograph. Arrange the four people from left to right; no person appears twice.



Available: **A, B, C, D, E, F.**

Choose B for the leftmost position.

Four positions, six people

Choose four of A, B, C, D, E, and F for a photograph. Arrange the four people from left to right; no person appears twice.



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Choose C for the second position.

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Choose four of A, B, C, D, E, and F for a photograph. Arrange the four people from left to right; no person appears twice.



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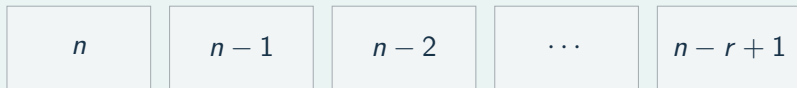
E and F are left out. By the product rule,

$$P(6, 4) = 6 \cdot 5 \cdot 4 \cdot 3 = 360.$$

Permutations

MUST UNDERSTAND A permutation is an ordered arrangement of distinct objects.

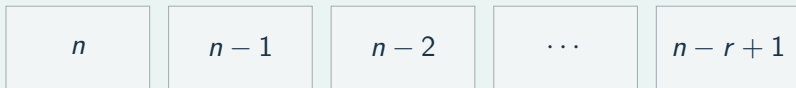
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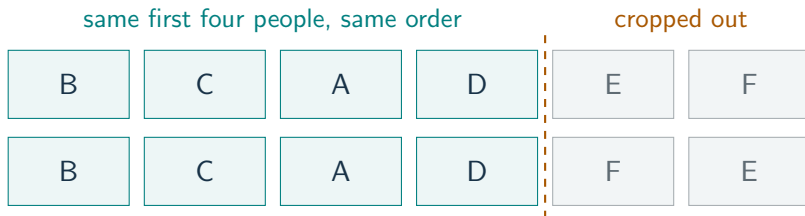


$$P(n, r) = n(n - 1) \cdots (n - r + 1)$$

Permutations using the division rule

Arrange all six people, then **crop the photograph to the first four**.

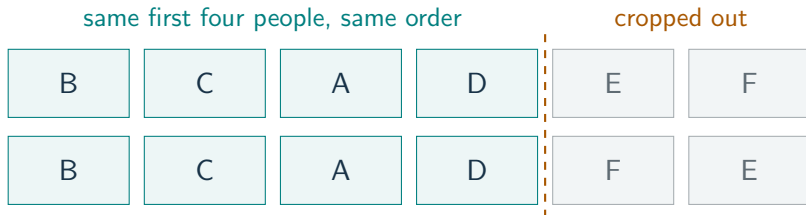
How many full photographs give the same cropped photograph?



Permutations using the division rule

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How many full photographs give the same cropped photograph?



Each cropped photograph comes from exactly $2!$ full photographs.

$$P(6, 4) = \frac{6!}{2!} = \frac{720}{2} = \boxed{360}.$$

The permutation formula

Arrange all n people, then keep only the first r positions.

first r positions: fixed

the photograph we keep

remaining $n - r$ people

$(n - r)!$ possible orders

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Every cropped photograph is counted $(n - r)!$ times among the $n!$ full arrangements.
By the division rule,

$$P(n, r) = \frac{n!}{(n - r)!}.$$

Example 1: Awards

Eight finalists compete for gold, silver, and bronze, with no ties. How many assignments of the three medals are possible?

Gold: 8

Silver: 7

Bronze: 6

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$$P(8, 3) = 8 \cdot 7 \cdot 6 = \boxed{336}.$$

Example 2: A fixed block

Count arrangements of ABCDEFGH that contain ABC consecutively in this order.

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The block is one object, so we arrange six objects:

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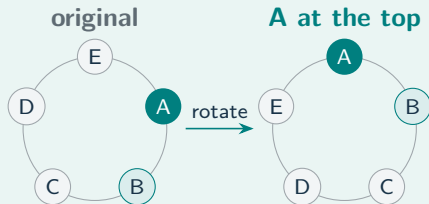
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$$6! = 720.$$

If the three letters may appear in any order inside the block, multiply by $3!$.

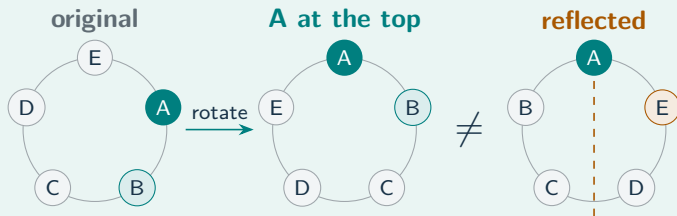
Circular arrangements

Seat $n \geq 1$ distinct people around a table with unlabelled seats. Rotations count as the same seating; reflections are not identified.



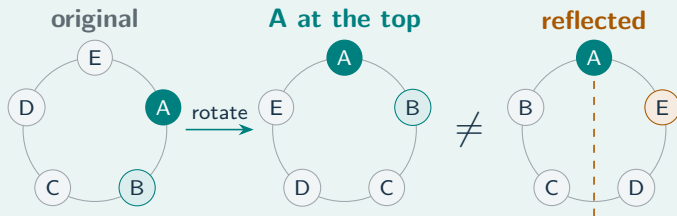
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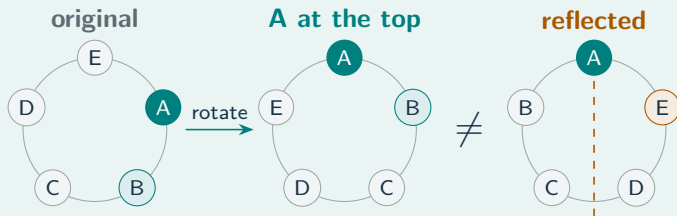
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Always rotate the arrangement so that A is at the top.

Fix A, then permute the remaining $n - 1$ people clockwise: $(n - 1)!$.

SECTION 3

Combinations

Selecting three people from five

Choose a three-person team from A, B, C, D, and E. Only the selected people matter; the team is an unordered subset.

How many different teams can we form?

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$$P(5, 3) = 5 \cdot 4 \cdot 3 = 60.$$

But (A, B, C) and (B, A, C) describe the **same team**.

One team, six ordered descriptions

The team $\{A, B, C\}$ appears in six orders:

A, B, C A, C, B B, A, C
B, C, A C, A, B C, B, A

One team, six ordered descriptions

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A, B, C A, C, B B, A, C
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Every three-person team is counted exactly $3! = 6$ times.

Combinations using the division rule

Division rule: if every outcome is counted exactly X times, divide the count by X .

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For three-person teams selected from five people:

$$\frac{\overbrace{P(5, 3)}^{60 \text{ ordered selections}}}{\underbrace{3!}_{6 \text{ orders per team}}} = \frac{60}{6} = \boxed{10 \text{ teams}}.$$

Combinations

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Read $\binom{n}{r}$ as “ n choose r ”.

Example: A five-country council

The United Nations has **193 member states**.

Suppose we select five countries to form a hypothetical advisory council. Any five distinct members may serve; all five have the same role.



How many different councils can we select?

Source: United Nations, Main Bodies.

Five-country council: Calculation

1. Count ordered selections of five distinct countries.

$$P(193, 5) = 193 \cdot 192 \cdot 191 \cdot 190 \cdot 189.$$

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2,118,000,528 possible councils.

Example: School IDs

A school ID has exactly five characters.

- Each character is an uppercase letter A–Z or a digit 0–9.
- An ID contains **at most two letters**.
- Characters may repeat; leading zeroes are allowed.

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How many different IDs are possible?

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How many different IDs are possible?

By the sum rule, split into three disjoint cases:

0 letters

1 letter

2 letters.

School IDs: Zero or one letter

Case 1: No letters. All five positions contain digits:

$$10^5 = 100,000.$$

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Case 1: No letters. All five positions contain digits:

$$10^5 = 100,000.$$

Case 2: Exactly one letter.

Choose its position, fill it with a letter, then fill the other four with digits:

$$\underbrace{\binom{5}{1}}_{\text{letter position}} \cdot \underbrace{26}_{\text{letter}} \cdot \underbrace{10^4}_{\text{four digits}} = 1,300,000.$$

School IDs: Two letter positions

Case 3: Exactly two letters. First choose which two of the five positions contain letters.



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$$C(5, 2) = \binom{5}{2} = \frac{5 \cdot 4}{2!} = \boxed{10 \text{ position choices}}.$$

School IDs: Fill the positions

For each choice of two letter positions:

- Each letter position has 26 choices.
- Each of the other three positions has 10 choices.

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By the product rule,

$$\underbrace{\binom{5}{2}}_{\text{letter positions}} \cdot \underbrace{26^2}_{\text{two letters}} \cdot \underbrace{10^3}_{\text{three digits}} = \boxed{6,760,000}.$$

School IDs: Total count

Number of letters	Letter-position choices	Number of IDs
0	$\binom{5}{0} = 1$	$10^5 = 100,000$
1	$\binom{5}{1} = 5$	$5 \cdot 26 \cdot 10^4 = 1,300,000$
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By the sum rule,

$$100,000 + 1,300,000 + 6,760,000 = \boxed{8,160,000}.$$

Example: Choose 18 of 20 students

A class has 20 students. Choose 18 to go on a trip.

How many different groups can go?

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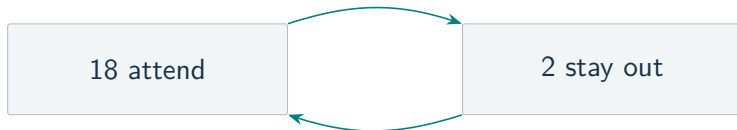
How many different groups can go?

Choose the 18 students who attend:

$$\binom{20}{18} = \frac{20!}{18! 2!} = \frac{20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}.$$

Example: Choose who stays out

Choosing the 18 students who attend uniquely determines the two who stay out.



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Choose the two who stay out instead of the eighteen who attend:

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Choose the two who stay out instead of the eighteen who attend:

$$\binom{20}{18} = \frac{20!}{18! 2!} = \frac{20!}{2! 18!} = \binom{20}{2}.$$

Expand only the factors that do not cancel:

$$\frac{20 \cdot 19 \cdot \cancel{18!}}{2 \cdot 1 \cdot \cancel{18!}} = \frac{20 \cdot 19}{2 \cdot 1} = \boxed{190}.$$

Complement identity

For integers $0 \leq r \leq n$,

$$\boxed{\binom{n}{r} = \binom{n}{n-r}}.$$

Choosing r objects to include is equivalent to choosing $n - r$ to leave out.

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The factorial formulas have the same denominator:

$$\binom{n}{n-r} = \frac{n!}{(n-r)! \underbrace{(n-(n-r))!}_{r!}} = \frac{n!}{(n-r)! r!} = \binom{n}{r}.$$

Subsets and bit strings

Encode a subset of $\{s_1, \dots, s_6\}$ by inclusion bits.

subset	bit string
$\{s_2, s_4, s_5\}$	0 1 0 1 1 0

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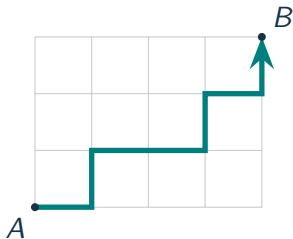
A size- r subset corresponds to a length- n bit string with r ones.

$$\#\{\text{length-}n \text{ strings with } r \text{ ones}\} = \binom{n}{r}.$$

Example: Paths on a grid

How many paths go from $A = (0, 0)$ to $B = (4, 3)$, using only unit steps **right** or **up**?

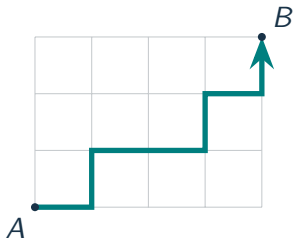
Every path has four R steps and three U steps.



position	1	2	3	4	5	6	7
step	R	U	R	R	U	R	U

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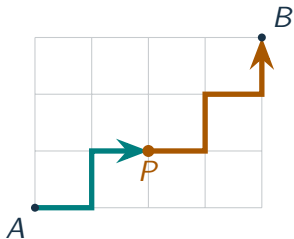
position	1	2	3	4	5	6	7
step	R	U	R	R	U	R	U

Choose the three U positions:

$$\binom{7}{3} = \boxed{35}.$$

Paths through a checkpoint

Now require the path to pass through $P = (2, 1)$.

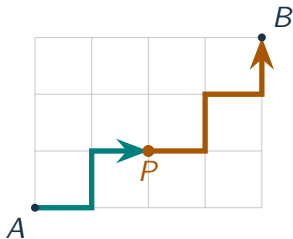


Split the path at P :

- $A \rightarrow P$: two R , one U .
- $P \rightarrow B$: two R , two U .

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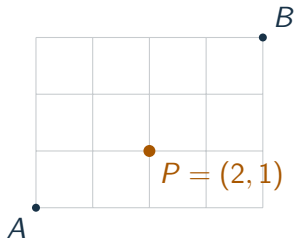
Split the path at P :

- $A \rightarrow P$: two R , one U .
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$$\underbrace{\begin{pmatrix} 3 \\ 1 \end{pmatrix}}_{A \rightarrow P} \cdot \underbrace{\begin{pmatrix} 4 \\ 2 \end{pmatrix}}_{P \rightarrow B} = \boxed{18}.$$

Exercise: Avoid the checkpoint

From $A = (0, 0)$ to $B = (4, 3)$, use only unit steps right or up.



How many paths do not visit P ?

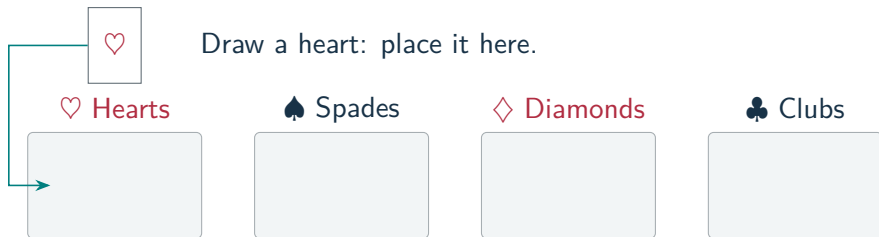
SECTION 4

Repetition

Example: Six cards, four suits

Draw six cards from a standard 52-card deck, without replacement.

After each draw, place the card in its suit's location. Ignore ranks and order.



Example: Six cards, four suits

Draw six cards from a standard 52-card deck, without replacement.

After each draw, place the card in its suit's location. Ignore ranks and order.

After six draws: one possible outcome



How many different final suit counts are possible?

Examples: Six cards sorted by suit

Three possible outcomes after sorting six cards by suit:



Counts: (, , , )

(6, 0, 0, 0)

(3, 0, 2, 1)

(1, 2, 1, 2)

Each row gives a different final count in the four locations.

Why dividing by $6!$ fails

There are 4^6 possible suit sequences. Can we simply divide by $6!$?

Suit counts	Number of suit sequences
Six hearts	1 : 

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The overcount is not uniform.

We cannot divide 4^6 by one common number of orders.

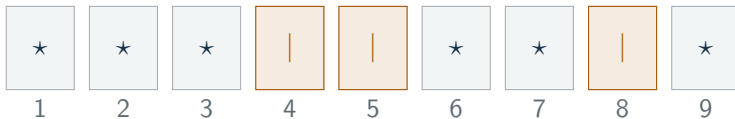
Stars and bars: From stars to suits

Six stars represent six cards. Three bars separate four suit groups.

Before bar 1: ♥.

Between bars 1 and 2: ♠.

Between bars 2 and 3: ♦. After bar 3: ♣.



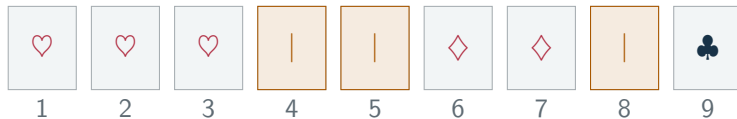
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Suit counts: (♥, ♠, ♦, ♣) = (3, 0, 2, 1).

Stars and bars: More examples

Keep the suit order fixed: ♥, ♠, ♦, ♣.

♥	♠	♦	♣		Pattern
3	0	2	1	↔	*** ** *

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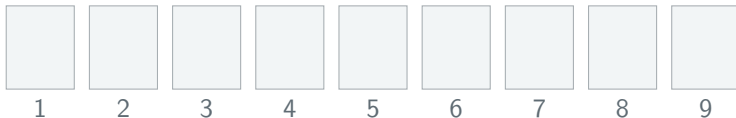
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1	2	1	2	↔	* ** * **
6	0	0	0	↔	*****

Bars may touch or sit at an end: some suit counts can be zero.

Stars and bars: Count the patterns

Nine positions: six stars and three bars.



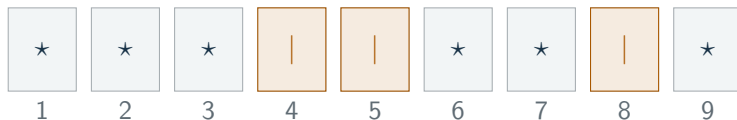
Number of four-suit count lists for six cards

=

Number of placements of $(4 - 1)$ bars in $(4 - 1) + 6$ slots

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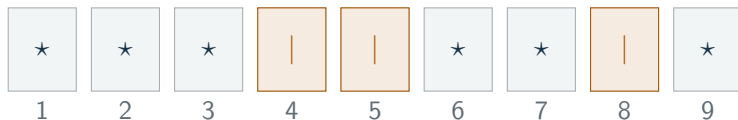
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$$\binom{9}{3} = \frac{9 \cdot 8 \cdot 7}{3 \cdot 2 \cdot 1} = \boxed{84 \text{ suit-count outcomes}}.$$

Combinations with repetition

MUST UNDERSTAND Choose r objects from n types; order is ignored.

Each type has unlimited supply. Here $n \geq 1$ and $r \geq 0$ are integers.

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Choose the $n - 1$ bar positions; the remaining slots are stars:

$$\boxed{\binom{r + n - 1}{n - 1} = \binom{r + n - 1}{r}}.$$

Stars and bars: At least one of each type

Choose r objects from $n \geq 1$ types, each with unlimited supply.

What if every type must appear?

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Distribute the remaining $r - n$ objects with zero amounts allowed.

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$$\binom{(r - n) + (n - 1)}{n - 1} = \boxed{\binom{r - 1}{n - 1}} \quad (r \geq n).$$

If $r < n$, there are no possible selections.

Example 4: Nondecreasing digits

Count six-digit numbers whose digits never decrease.

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2. Order is forced: $1, 1, 3, 3, 3, 8 \longleftrightarrow 113338$.

3. Count the multiset:

$$\boxed{\binom{9+6-1}{6} = \binom{14}{6}}.$$

SECTION 5

Problem solving

Choose the counting model

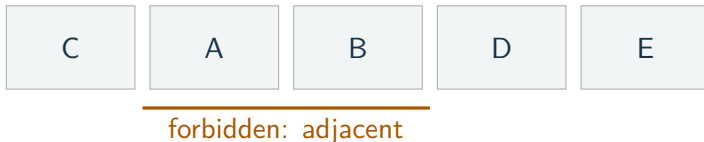
	Order matters	Order ignored
No repetition	Arrange r of n people in a photograph. $P(n, r) = \frac{n!}{(n - r)!}$	Select a team of r from n people. $\binom{n}{r}$
Repetition allowed	Form a length- r code using n symbols. n^r	Choose r items from n types. $\binom{n + r - 1}{n - 1}$

No repetition: $0 \leq r \leq n$. Repetition: $n \geq 1, r \geq 0$, unlimited supply.

Challenge: A group photograph

Eight distinct students include A and B.

- Choose five students and arrange them in a line.
- A and B must both be included.
- They must **not stand next to each other**.



How many different photographs are possible?

Roadmap: Choose, arrange, subtract

1. **Choose the other three members.**

A and B are already required.

2. **Count all orders of that five-person group.**

3. **Subtract orders with A and B together.**

Treat the adjacent pair as one block.

Group photograph: Count the valid orders

Choose the other three students: $\binom{6}{3} = 20$ groups.

All orders of one group



$$5! = 120$$

Forbidden: adjacent pair



$$2 \cdot 4! = 48$$

Four objects; AB or BA inside the block.

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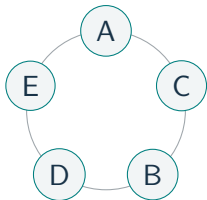
$$2 \cdot 4! = 48$$

Four objects; AB or BA inside the block.

$$\binom{6}{3} (5! - 2 \cdot 4!) = 20(120 - 48) = \boxed{1440}.$$

Exercise: Seat the group around a table

Choose five of the same eight students, including A and B.



Seat them around a round table with unlabelled seats.

- Rotations count as the same seating.
- Reflected arrangements count as different seatings.
- A and B may not be neighbours.

How many seatings are possible?

Practice

From seven women and six men, form a five-person committee with exactly three women. The chair must be one of the selected men.

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Choose the members:

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Choose one of the two selected men as chair:

$$\binom{7}{3} \binom{6}{2} \cdot 2 = \boxed{1050}.$$

Optional: A queue at the ticket counter

A ticket costs \$5. The cashier starts with **no change**.

- Ten customers each pay with a \$5 bill.
- Ten customers each pay with a \$10 bill and need \$5 change.

How many queues let everyone buy a ticket in order?

Count only the sequence of bill types, not the customers' identities.

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Equivalent validity condition: in every prefix (the first k customers),

$$\# \text{ customers paying } \$5 \geq \# \text{ customers paying } \$10.$$

This ensures that the cashier always has enough \$5 bills for change.

Two smaller queues

Start with three \$5 bills and three \$10 bills.

Successful queue	\$5	\$10	\$5	\$5	\$10	\$10
\$5 bills left	1	0	1	2	1	0

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\$5 bills left	1	0	cannot serve				

The third customer needs change, but the cashier has none.

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The third customer needs change, but the cashier has none.

In every initial segment: number of \$5 bills $\geq$ number of \$10 bills.

Draw a queue as a path

\$5 bill: **one step up.** \$10 bill: **one step down.**

Height = number of \$5 bills minus number of \$10 bills so far.



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Below zero means the queue would require unavailable change.

Count paths with ten up-steps and ten down-steps that never go below zero.

Count all queues first

Choose the ten positions occupied by \$5 bills.

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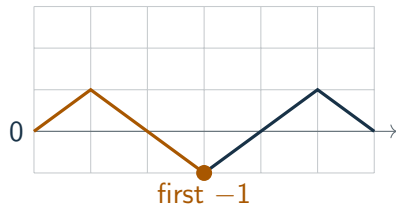
Successful queues = all queues – failed queues.

A failed path reaches height -1 at some point.

How can we count these paths without listing them?

Reflect up to the first failure

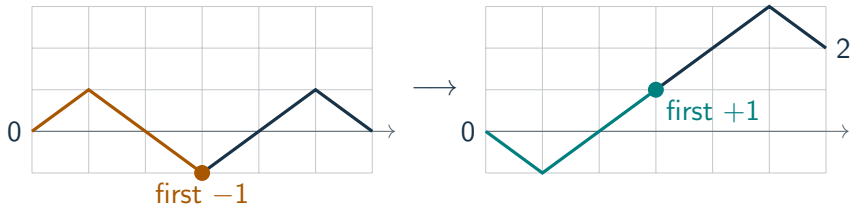
Reuse our failed queue: \$5, \$10, \$10, \$5, \$5, \$10.



1. Stop at the **first visit to -1**.

Reflect up to the first failure

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1. Stop at the **first visit to -1**.
2. Swap up and down in this initial segment.
3. Keep all later step directions unchanged. The endpoint rises to 2.

Count the steps after reflection

At the first failure, the initial segment has one extra down-step.

Initial segment	Up-steps	Down-steps
Before reflection	a	$a + 1$

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The whole path gains one up-step and loses one down-step.

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For our twenty-customer problem:

$$(10 \text{ up}, 10 \text{ down}) \longrightarrow \boxed{11 \text{ up}, 9 \text{ down}}.$$

Why each failed queue is counted once

Start with **any** path having eleven up-steps and nine down-steps.

It ends at height 2, so it must visit height $+1$.

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Reflect the initial segment ending at its **first visit to $+1$** .



We recover a failed path with ten up-steps and ten down-steps.

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It ends at height 2, so it must visit height +1.

Reflect the initial segment ending at its **first visit to +1**.



We recover a failed path with ten up-steps and ten down-steps.

Before the first +1, heights are ≤ 0 ; after reflection they are ≥ 0 .

The two operations undo each other: a one-to-one correspondence.

Count the failed queues

Failed paths with 10 up-steps and 10 down-steps

$\updownarrow$ reflection and its reverse

All paths with 11 up-steps and 9 down-steps

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All paths with 11 up-steps and 9 down-steps

Choose the nine down-step positions among twenty:

$$\text{Failed queues} = \binom{20}{9}.$$

There is no height restriction on these transformed paths.

The number of successful queues

$$\underbrace{\binom{20}{10}}_{\text{all queues}} - \underbrace{\binom{20}{9}}_{\text{failed queues}} = 184,756 - 167,960 = \boxed{16,796}.$$

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The key step: count the failed queues through a reversible transformation.

For n customers of each bill type, the same argument gives

$$\binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}.$$

This is the n th **Catalan number**.

Takeaways

- Use a permutation when order changes the outcome.
- Use a combination when only the selected subset matters.
- Separate membership choices from role assignments.
- Use blocks for objects that must stay together.
- Use stars and bars only when repetition is allowed and order is ignored.